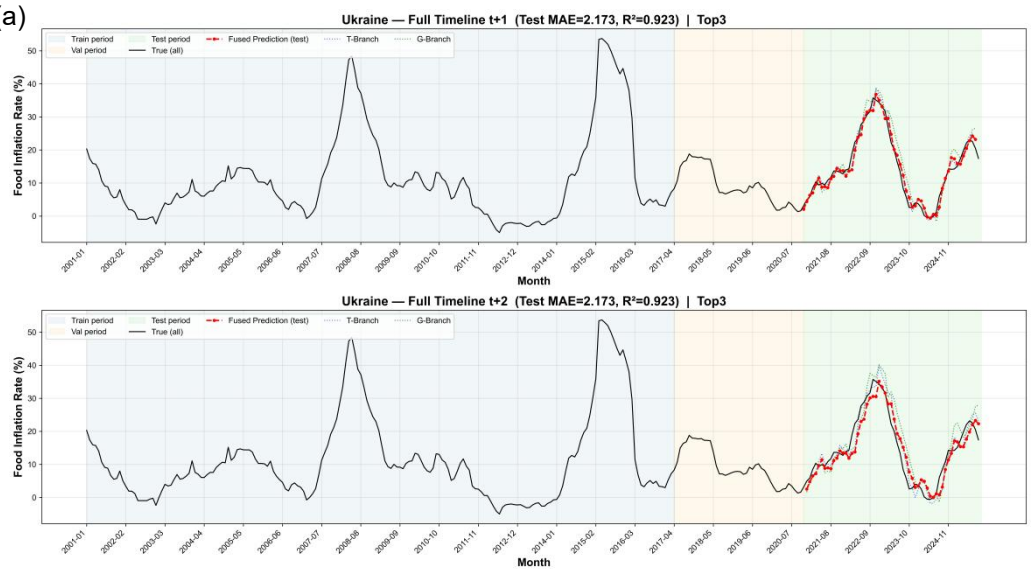
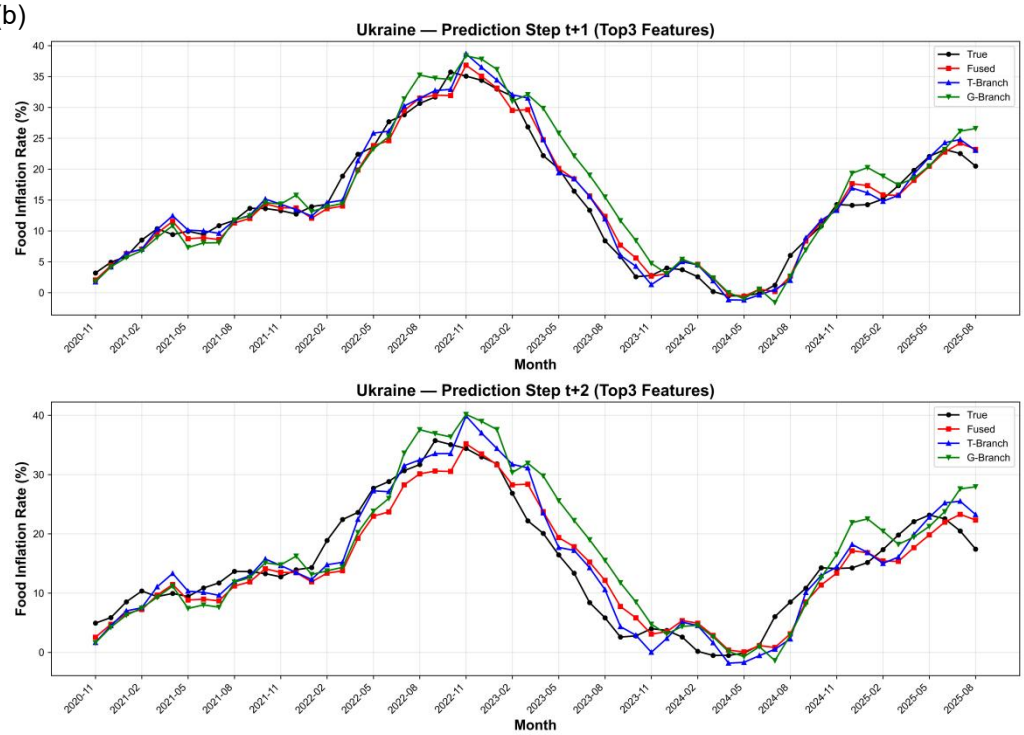
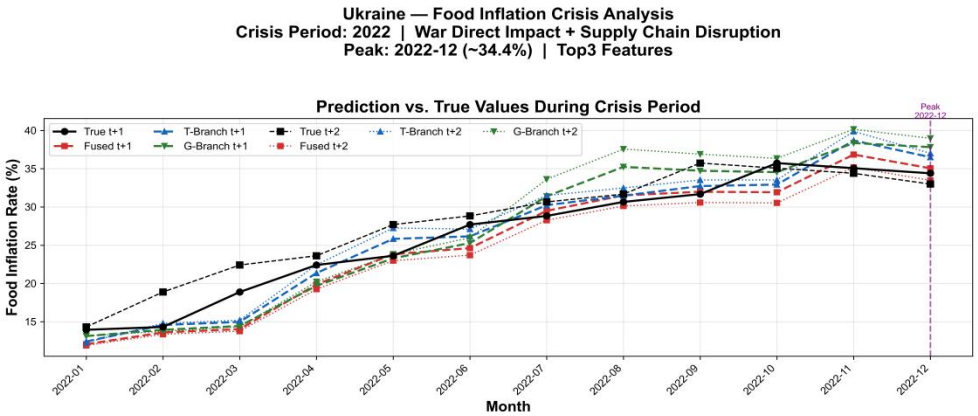
(a)


Ukraine — Full Timeline t+1 (Test MAE=2.173, R²=0.923) | Top3
Train period
Test period
Fused Prediction (test)
T-Branch
G-Branch
Val period
True (all)
Food Inflation Rate (%)
Month
Ukraine — Full Timeline t+2 (Test MAE=2.173, R²=0.923) | Top3
Train period
Test period
Fused Prediction (test)
T-Branch
G-Branch
Val period
True (all)
Food Inflation Rate (%)
Month

(b)


Ukraine — Prediction Step t+1 (Top3 Features)
True
Fused
T-Branch
G-Branch
Food Inflation Rate (%)
Month
Ukraine — Prediction Step t+2 (Top3 Features)
True
Fused
T-Branch
G-Branch
Food Inflation Rate (%)
Month

(c)


Ukraine — Food Inflation Crisis Analysis
Crisis Period: 2022 | War Direct Impact + Supply Chain Disruption
Peak: 2022-12 (~34.4%) | Top3 Features
Prediction vs. True Values During Crisis Period
Peak
2022-12
True t+1
Fused t+1
T-Branch t+1
G-Branch t+1
True t+2
Fused t+2
T-Branch t+2
G-Branch t+2
Food Inflation Rate (%)
Month

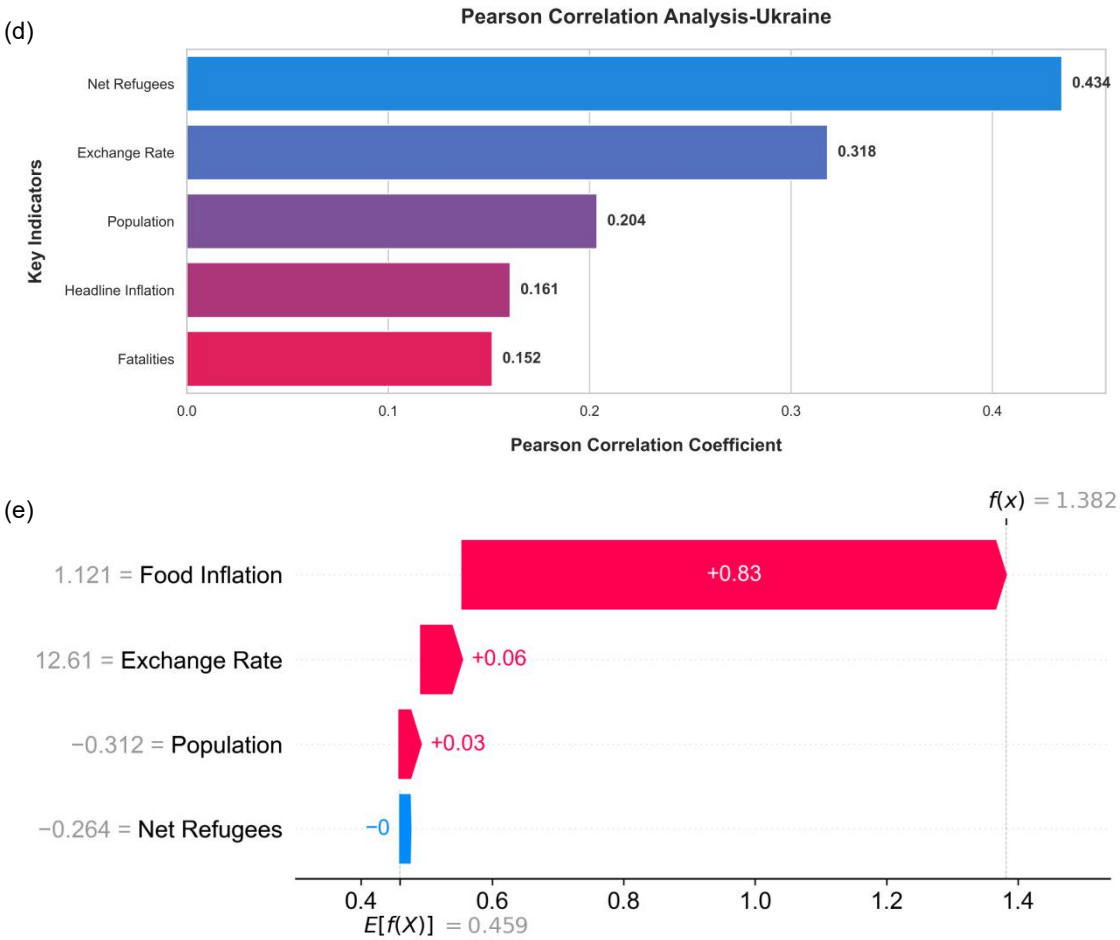



Figure : Multidimensional Evaluation and Attribution Analysis of Ukraine's Food Inflation Forecast.(a) Overview of the total cycle prediction and fitting, (b) Multi-step Short-Term Forecast Comparison Chart, (c) Details of Prediction Performance in Extreme Crisis Periods, (d) Pearson correlation analysis of macroeconomic drivers, (e) SHAP-based explanation of feature contributions for a specific prediction instance.

The frequent damage to the energy infrastructure and the reconfiguration of the logistics system caused by the war are the core contradictions behind the fluctuation of food prices in **Ukraine**. Although Ukraine is a globally important grain-producing country, its domestic raw material supply is relatively abundant when exports are blocked, but the power facilities being attacked has severely disrupted the continuity of food processing, cold chain storage, and bread production, forcing producers to invest in expensive generator equipment, resulting in a significant increase in costs. Under wartime conditions, traditional railway and port logistics have been replaced by costly road transportation. Fuel shortages and high insurance costs have further pushed up the cost of every kilometer from the field to the table. Moreover, the loss of labor force has led to a shortage of manpower in the harvesting and processing stages, and the salary premium has also been fixed in the prices. The food inflation in Ukraine presents a special balance: the price of raw materials is suppressed due to surplus, but the processing and distribution stages have become extremely expensive due to war damage.